

Product Carbon Footprint Study Report

Product Name: integrated circuit production, logic type

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: integrated circuit production, logic type

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 kg integrated circuit production, memory type

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 kg integrated circuit production, logic type.

2. System Boundary

integrated circuit production, logic type, in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of integrated circuit production, logic type produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **1619.681 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

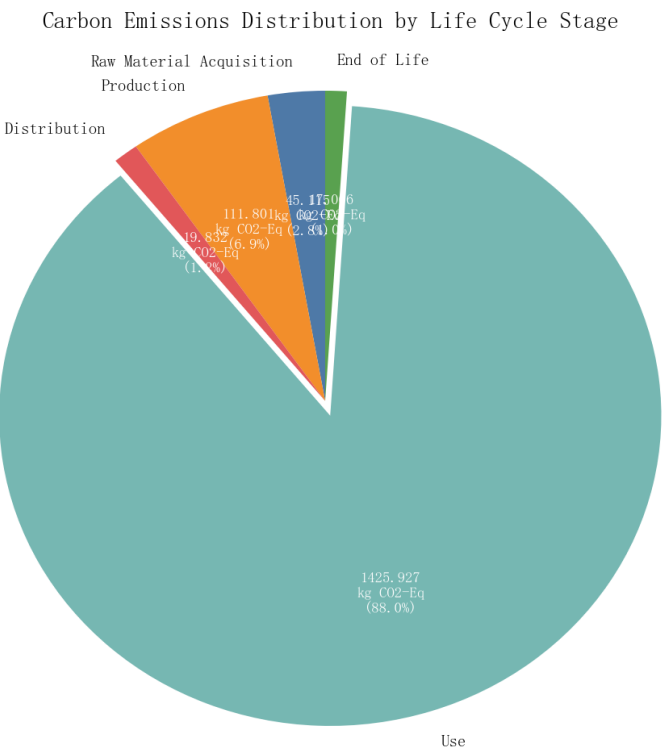


Table 1: Carbon Emissions by Life Cycle Stage

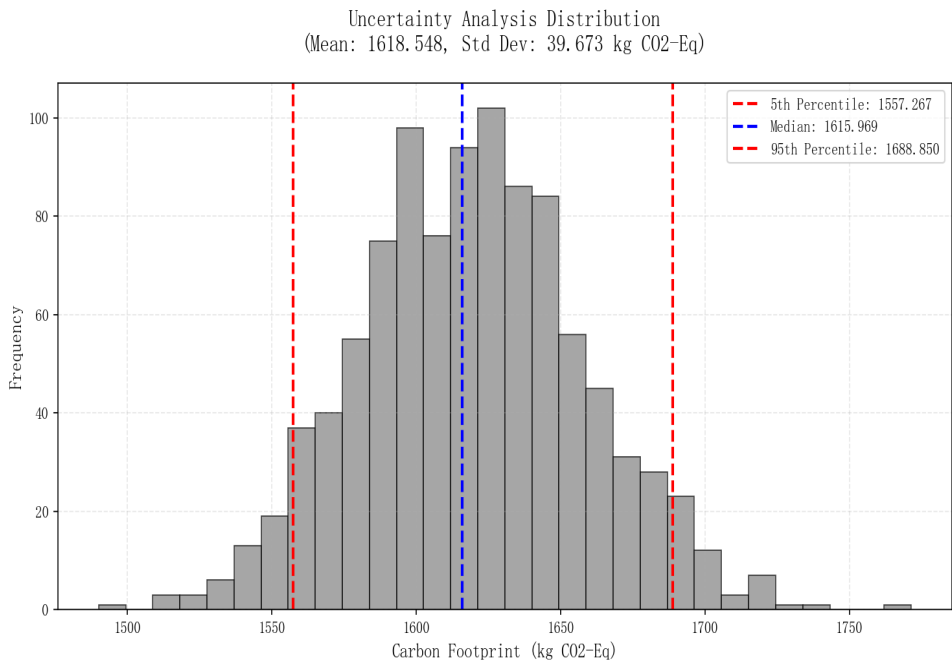
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	45.115	2.79
Production	111.801	6.90
Distribution	19.832	1.22

Use	1425.927	88.04
End of Life	17.006	1.05
Total	1619.681	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	1618.548
Standard Deviation	39.673
Coefficient of Variation (CV)	2.45%
Median	1615.969
5th Percentile	1557.267
95th Percentile	1688.850

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the integrated circuit (logic type) is calculated at 1619.681 kg CO₂-Eq per functional unit. This value is relatively high, reflecting the energy-intensive nature of semiconductor use, particularly in applications such as data centers or consumer electronics where the device operates for extended periods. The mean from the uncertainty analysis (1618.548 kg CO₂-Eq) aligns closely with the deterministic result, and the low coefficient of variation (2.45%) indicates good precision in the overall estimate.

Analysis of lifecycle stage contributions reveals a highly skewed distribution: the use phase dominates with 88.0% of total emissions (1425.927 kg CO₂-Eq), while raw material extraction (2.8%), production (6.9%), distribution (1.2%), and end-of-life (1.0%) are comparatively negligible. This pattern is typical for logic integrated circuits that consume significant electricity during operation, especially if powered continuously over a long lifetime. The production stage, though small relative to use, still contributes over 100 kg CO₂-Eq, which may be relevant for low-use applications.

The major emission source is unequivocally the use phase, driven by electricity consumption. The top process contributions are all listed as “Unknown” (0.000e+00 kg CO₂-Eq), indicating that the underlying process-level data were not resolved or were aggregated. This limitation prevents identification of specific manufacturing or supply chain hotspots, but the lifecycle stage breakdown clearly prioritizes the use phase as the primary target for reduction.

2. Statement of Assumptions and Limitations

Key assumptions in this study include a specific use-phase scenario: likely an assumed operational lifetime (e.g., 10–15 years) and a constant power draw typical for logic ICs in server or high-performance applications. The electricity grid mix used for the use phase is not specified but is probably based on a global or regional average. The production and raw material stages assume typical semiconductor fabrication processes (e.g., CMOS) with average yields and material inputs. The distribution stage likely uses a generic transport mode and distance.

Limitations arise from the data quality: the “Unknown” entries for top processes suggest that the LCA model relied on aggregated datasets (e.g., from ecoinvent or similar) without transparent process-level contributions. The coefficient of variation (2.45%) is low, but this may reflect limited variability in the input parameters rather than high data accuracy. The 90% confidence interval (1557–1689 kg CO₂-Eq) provides a range, yet the actual uncertainty could be larger if key assumptions (e.g., use-phase electricity mix) are varied. The study lacks explicit information on the specific IC type, wafer size, or fabrication node, which are critical for production emissions.

System boundary restrictions include a cradle-to-grave scope (raw material to end-of-life), but the end-of-life stage only accounts for 1.0%, likely assuming a landfill or recycling scenario with minimal emissions. The use phase boundary may not include auxiliary components (e.g., cooling in a data center). Additionally, infrastructure (e.g., fab construction) and capital goods are excluded, which could increase the production stage contribution.

3. Suggestions for Improvement

- ****Target the use phase****: Implement energy-efficient design for the integrated circuit, such as reducing operating voltage or adopting advanced low-power architectures (e.g., near-threshold computing). For applications where the IC is used continuously, prioritize power management features that allow sleep modes or dynamic voltage scaling.
- ****Optimize production energy****: Improve the energy efficiency of wafer fabrication by using renewable electricity sources (solar, wind) in manufacturing facilities. This can reduce the production stage's carbon intensity, which, while currently small, becomes significant for high-volume chips.
- ****Enhance data quality****: Replace "Unknown" process contributions with specific data from the IC manufacturer. Collect primary data on fabrication energy, chemical usage, and transport distances. Use actual use-phase electricity mix data (regional or customer-specific) to refine the dominant contribution.
- ****Consider lifetime and application****: For ICs used in short-lifetime products (e.g., smartphones), the use-phase impact may be lower; adjust the functional unit or scenario accordingly. Perform sensitivity analysis on use-phase duration and power consumption to identify reduction opportunities.
- ****Improve end-of-life modeling****: Investigate recycling or refurbishment pathways for logic ICs to reduce the small but non-zero end-of-life emissions. Include potential benefits from material recovery (e.g., gold, silicon) in future assessments.